BITEHUB

Restaurant Ordering System

CSC1426 — Advanced Web Programming
Project Phase 1

Student name: _____

Date: _____

Individual Project

Abstract

BiteHub is a responsive frontend restaurant ordering website. Visitors can browse food categories, search dishes, view dynamic dish details, add meals to a basket, change quantities, and review an automatically calculated total. The project demonstrates component-based web development and client-side interaction with React.

1. Project Description

The purpose of BiteHub is to provide a simple and attractive digital menu and ordering experience. It is designed for desktop and mobile devices. The project is a frontend prototype and does not process real payments or send orders to a restaurant server.

2. System Design

Home: restaurant introduction and featured meals.
About: restaurant story and values.
Menu: food cards, search, and category filters.
Dynamic Details: individual dish page using a route parameter.
Order: cart, quantity updates, total calculation, and demo checkout form.
Contact: restaurant information and contact form.

3. Technologies Used

React builds reusable interface components and manages application state. Vite provides the development server and production build. React Router handles navigation and dynamic routes. CSS3 is used for styling and responsive layouts.

4. Functional Requirements

Users can search for meals and filter by category. They can add items to the cart, update quantities, remove items, and view the total. The demo checkout validates required fields before showing a confirmation. The contact form displays a confirmation after submission.

5. Project Structure

```
BiteHub_Restaurant_Project/\n  index.html\n  package.json\n  README.md\n  src/\n    main.jsx\n    App.jsx\n
```

6. Setup Instructions

Install Node.js LTS, extract the project folder, open a terminal in the folder, then execute:

```
npm install\nnpm run dev\nnpm run build
```

7. Code Snippet — Cart State

```
function add(dish) {\n  setCart(current => current.some(x => x.id === dish.id)\n    ? current.map(x => x.id ===
```

8. Conclusion

BiteHub demonstrates a multi-page React frontend with responsive design, dynamic routing, filtering, search, and interactive cart functionality. Future work could connect the frontend to a backend API, database, authentication, and a payment service.

9. Screenshots

Capture screenshots of the running Home, Menu, and Order pages and add them to the final report if requested by the instructor.